

Intel is also building Lake Crest (or Nervana Engine), a full-stack solution for deep learning, optimised at every stage to make training ten times faster.

The second stage, or the inference, is undergoing a major shakeup. In the past, most of this work was also done on the cloud using GPUs, TPUs and FPGAs. However, now smart device makers are beginning to feel that this is not 'real-time' enough. So, the trend is turning in favour of implementing the inference or execution stage at the edge (that is, in the device itself) instead of relying on the cloud. Since existing devices don't have enough processing power, memory size and bandwidth for this, the semiconductor industry is facing a heavy demand for high-performance, low-power inference engines for deep neural networks, which can be built into devices.

AI goes local

In a press event held in San Francisco in March this year, Deepu Talla, vice president and general manager of Nvidia's Tegra business unit, cited four reasons in favour of bringing AI technology to the edge (on-board the device)—bandwidth, latency, privacy and availability. As the number of devices communicating with the cloud rises by the day, there will be a dire shortage of bandwidth in the near future. Latency is also an issue in applications like self-driving cars where a split second's delay can have serious implications. Privacy, of course, has always been a serious issue. So, when data is processed within the device itself, there is no lingering doubt about misuse on the cloud. Availability of the cloud is also questionable in rural areas where connectivity is unreliable. When AI is implemented at the

edge, all these issues are sorted out.

"We will see AI transferring to the edge," Talla said to the press, with future intelligent applications using a combination of edge and cloud processing.

This requirement to build intelligence into the device is leading to a major bustle in the semiconductor industry, not to forget a lot of hardware innovation. In last month's story on smart robotics, we read about a micromote (a chip measuring just one cubic millimetre) developed at the University of

be used in remote farms or even by children playing hide-and-seek! EnRoute, another drone maker, has used on-board AI to help its drones navigate and fly faster, avoiding objects on their path.

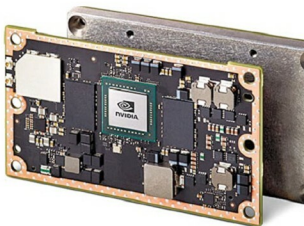
Cisco has developed a collaboration device that uses AI to recognise people in a room and automatically pick a field-of-view (FOV) with people in it instead of empty chairs. The FOV is spontaneously adjusted as people walk in and out, or move around. The system also zooms in on people who are speaking.

Live Planet's new 360-degree 3D camera for live streaming of video uses on-board AI to encode 3D videos in real time. Live Planet's chief strategy officer Khayyam Wakil explains, "The camera produces a stream of 65 gigabytes, which is too much data to transmit to a cloud server. On-board processing has made the live streaming possible."

Sniffing the trend, Intel acquired Movius in 2016. Movius produces specialised low-power processor

chips for computer vision and deep learning. Their button-sized Myriad 2 platform has many features that support implementation of deep learning at the network edge. Myriad's SHAVE processor engines achieve the hundreds of giga-flops required in fundamental matrix multiplication compute that is essential for deep learning networks. The on-chip RAM keeps huge volumes of intermediate data on the chip itself to avoid bandwidth bottlenecks. The platform comes with native support for mixed precision and hardware flexibility—both 16-bit and 32-bit floating-point data types, as well as u8 and unorm8 types are supported.

The company literature explains that existing hardware accelerators can be easily repurposed to provide



Nvidia Jetson TX2 credit-card sized platform for intelligent edge devices like robots, drones, cameras and portable medical devices (Courtesy: Nvidia)

Michigan, which incorporates a deep-learning processor capable of operating a neural network using just 288 microwatts.

Last year, Nvidia launched Drive PX2—a palm-sized platform to implement auto cruise capabilities in automobiles. This open AI car platform features a unified architecture that allows deep neural networks to be trained on a system in the data centre, and then deployed in the car. This year, Nvidia launched Jetson TX2—a credit-card sized, plug-in edge-processing platform designed for embedded computing. Teal Drones has used the Jetson module to develop a smart drone that can understand and react to what its cameras are seeing. Since this drone does not rely on the cloud, it can